第五章 刚体的定轴转动

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§ 5.2 刚体定轴转动定律

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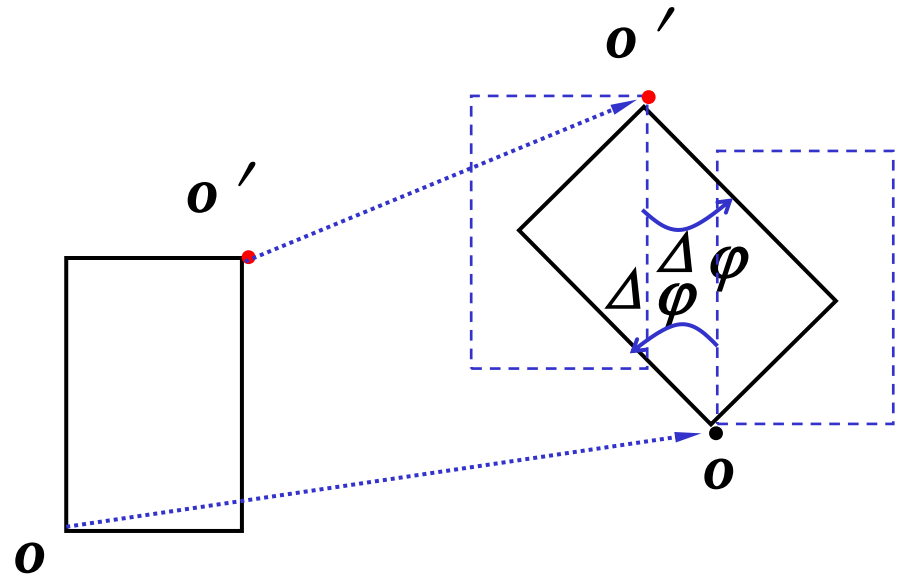
§ 5.1 刚体的运动

刚体（**rigid body**）：特殊的质点系，形状和体积不变化，理想化的模型。

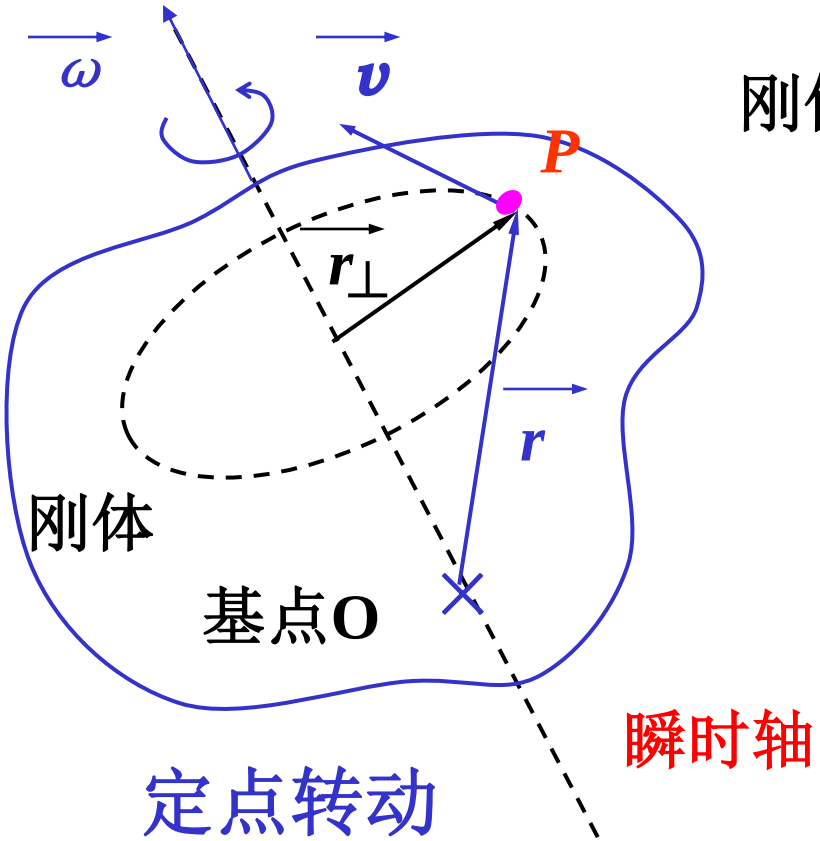
模型的适用原则

平动（**translation**）时，刚体上所有点运动都相同。



刚体质点间的相对运动只能是绕某一轴转动（**rotation**）的结果。

平动和转动，可以描述所有质元（质点）的运动 (欧拉定理)。



定点转动

刚体绕O的转动其转轴是可以改变的

反映瞬时轴的方向及转动快慢，
引入角速度矢量 $\vec{\omega}$
和角加速度矢量 $\vec{\alpha}$

$$\vec{\alpha} = \frac{d\vec{\omega}}{dt}$$

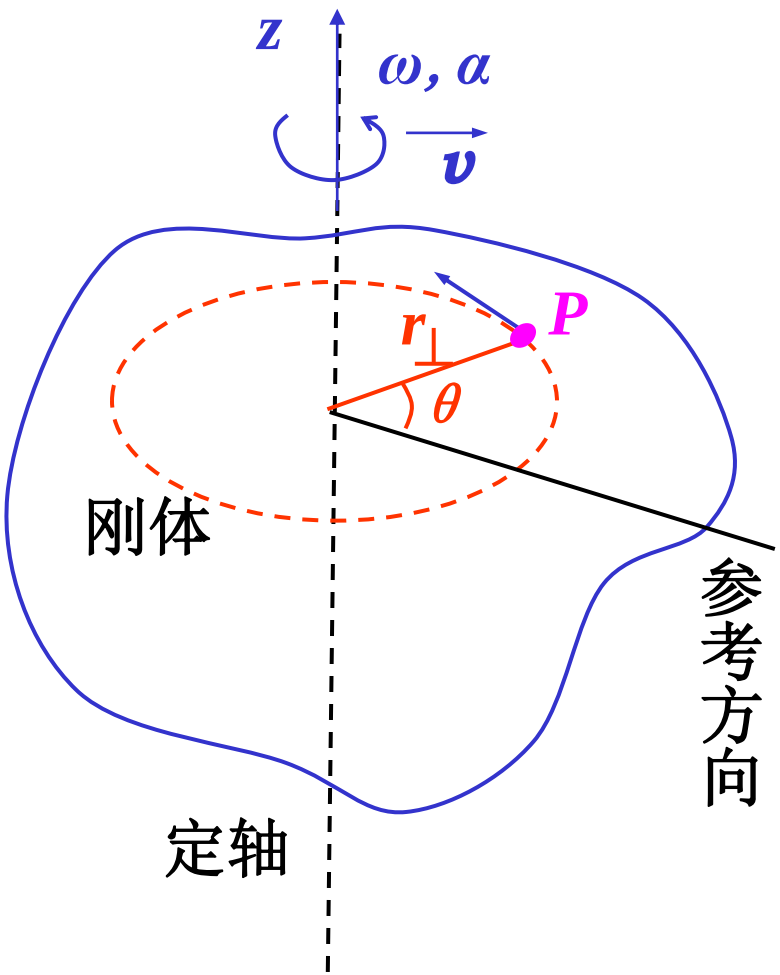
P点线速度

$$\vec{\upsilon} = \vec{\omega} \times \vec{r}_{\perp} = \vec{\omega} \times \vec{r}$$

P点的加速度

旋转加速度　　向轴加速度

$$\vec{a} = \frac{d\vec{\upsilon}}{dt} = \frac{d\vec{\omega}}{dt} \times \vec{r} + \vec{\omega} \times \frac{d\vec{r}}{dt} = \vec{\alpha} \times \vec{r} + \vec{\omega} \times (\vec{\omega} \times \vec{r})$$



定轴转动（rotation about affixed axis）：

$\vec{\omega},\vec{\alpha}$ 退化为代数量，刚体上任意点都绕同一轴作圆周运动，且 ω，α 都相同。

$$v = r_{\perp}\omega$$

$$a_n = r_{\perp}\omega^2,$$

$$a_t = \frac{dv}{dt} = r_{\perp}\alpha$$

定轴匀角加速转动

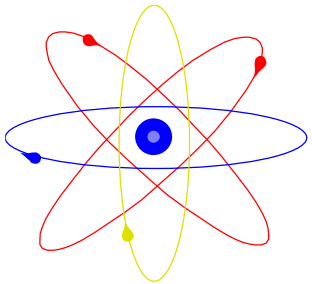
$$\alpha = const.$$

类比匀加速直线运动

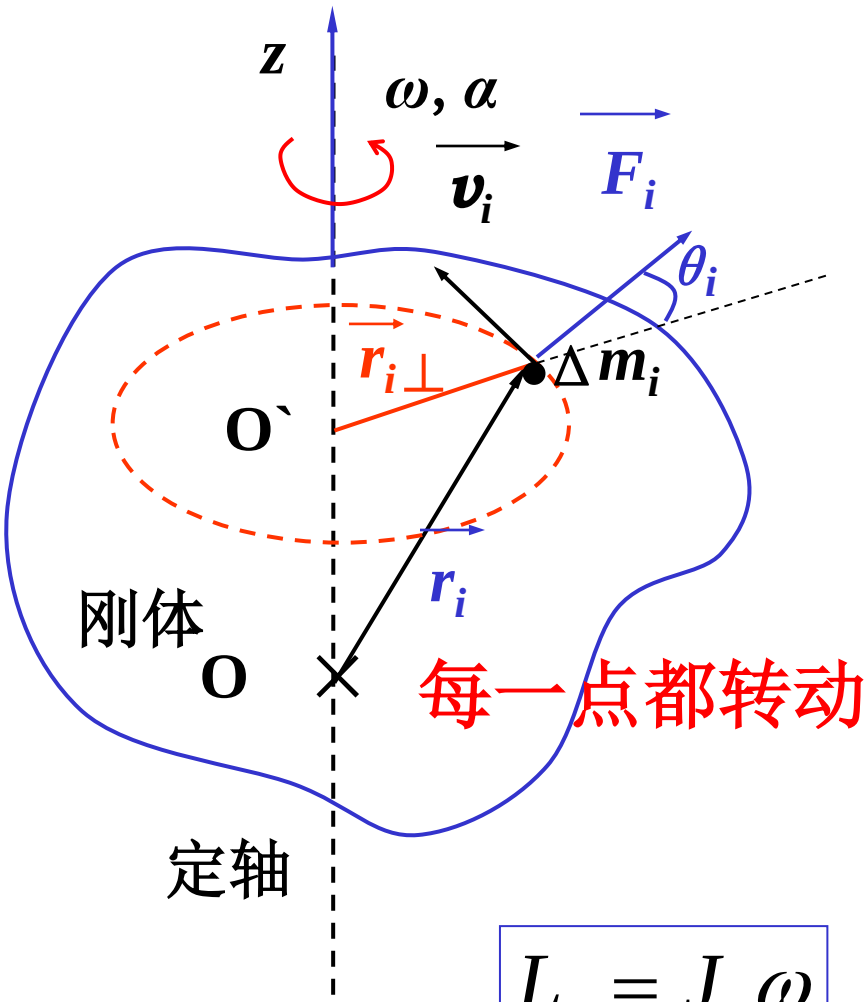
$$\alpha = \frac{d\omega}{dt}, \qquad \omega = \frac{d\theta}{dt} \longrightarrow a = \frac{d\upsilon}{dt}, \quad \upsilon = \frac{ds}{dt}$$

$$a \leftrightarrow \alpha, \quad \upsilon \leftrightarrow \omega$$

$$\begin{cases} \omega = \omega_0 + \alpha\, t \\ (\theta - \theta_0) = \omega\, t + \frac{1}{2}\alpha\, t^2 \\ \omega^2 - \omega_0^2 = 2\alpha(\theta - \theta_0) \end{cases}$$



§ 5.2 刚体的定轴转动定律



$$\boxed{L_z = J_z\omega}$$

类似于多质点系

$$\vec{M}_{外} = \frac{d\vec{L}}{dt} \qquad (对o点)$$

$$M_{外z} = \frac{dL_z}{dt} \qquad (对z轴)$$

$$L_z = \sum_i L_{iz} = \sum_i \Delta m_i v_i r_{i\perp}$$

$$= (\sum_i \Delta m_i r_{i\perp}^2)\cdot\omega$$

$$J_z = \sum_i \Delta m_i r_{i\perp}^2$$

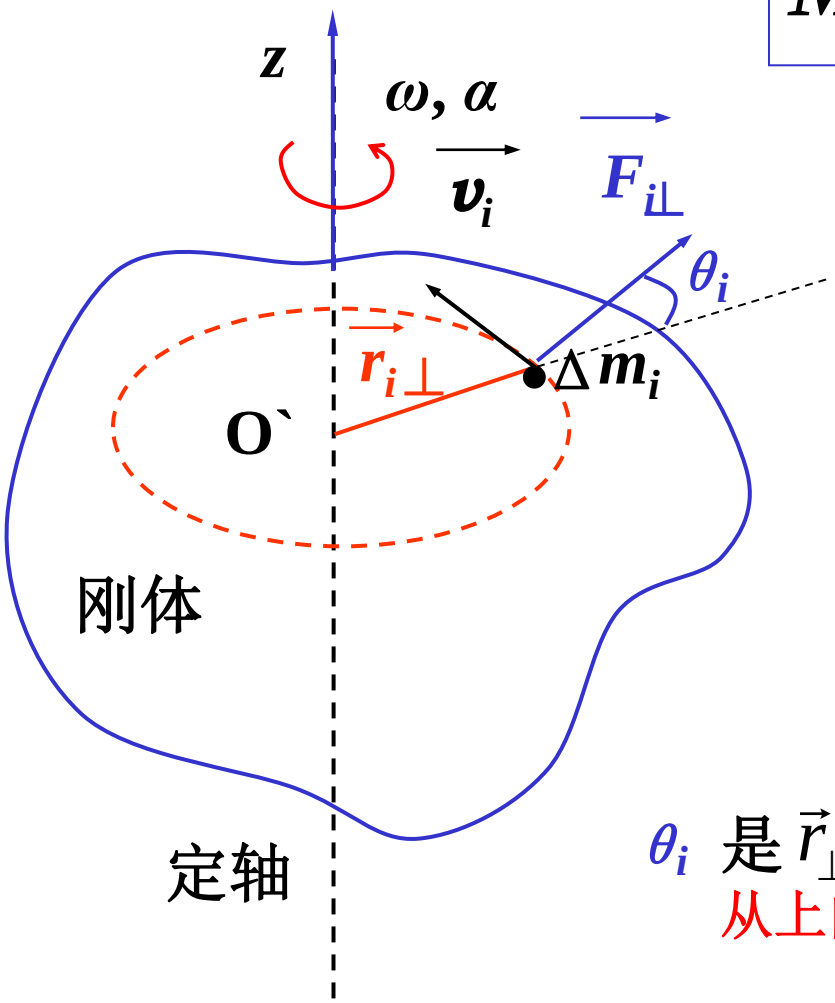
rotational inertia

J_z 称为刚体对 z 轴的转动惯量

转动惯量 J_z 是关于质量对z 轴的分布

$$M_{外z} = \frac{dL_z}{dt} = J_z \frac{d\omega}{dt}$$

$$\boxed{M_{外z} = J_z \alpha}$$ —— 转动定律



其中 $M_{外z} = \sum (\vec{r}_i \times \vec{F}_i) \cdot \hat{z}$

$$= \sum (\vec{r}_{i\perp} \times \vec{F}_{i\perp}) \cdot \hat{z}$$

$$M_{外z} = \sum_i F_{i\perp} r_{i\perp} \sin\theta_i$$

是对 z 轴的外力矩和。

θ_i 是 $\vec{r}_\perp$ 转到 $\vec{F}_\perp$ 的角度,

从上向下看(逆着 z 轴)，逆时针转规定为正向

定轴下，可不写角标 z，记作：

$$\boxed{M = J\alpha}$$

与牛II比较：

$$\begin{cases} M \sim F \\ J \sim m \\ \alpha \sim a \end{cases}$$

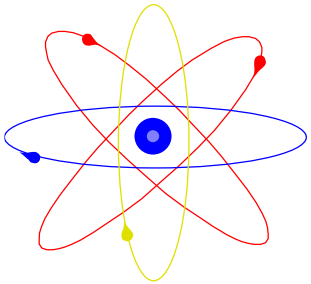
$\therefore$ J反映刚体转动的惯性

定轴转动与一维质点运动数学表示形式上一致

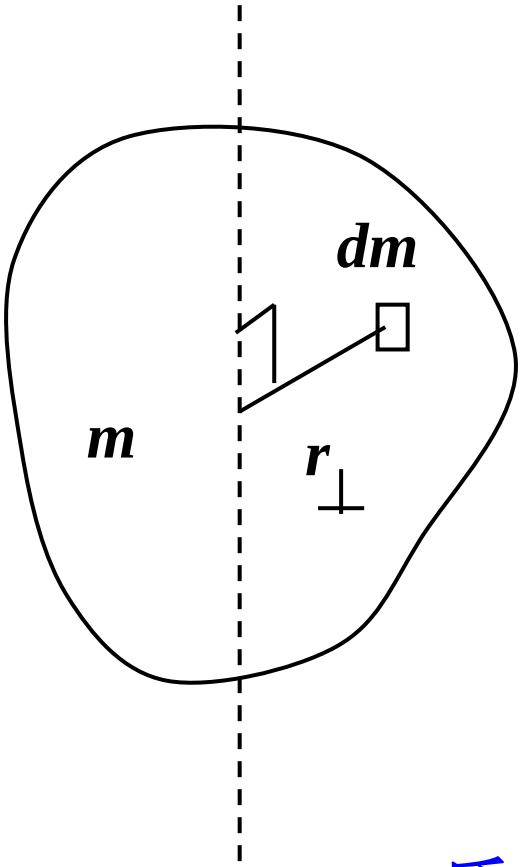
类比法的基础:

相同的方程具有相同的解

方程中的变量物理意义不同



§ 5.3 转动惯量的计算



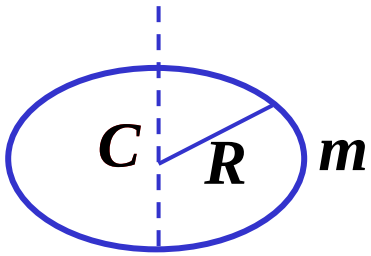
$$J = \sum \Delta m_i r_{i\perp}^2 \qquad \text{(分立)}$$

$$J = \int_m r_\perp^2 \cdot dm \qquad \text{(连续)}$$

J由质量对轴的分布决定。

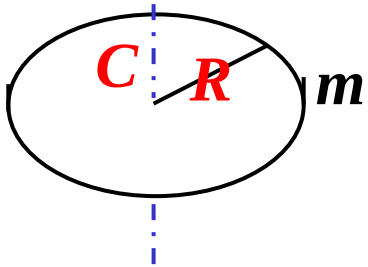
质点的定轴转动惯量: $J = mr_\perp^2$

一. 常用的几个J



均匀圆环： $J_C = mR^2$；

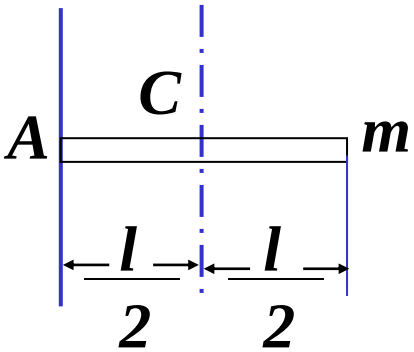
均匀圆盘：



$$J_c = \sum_i \Delta m_i r_i^2 = \sum_i \sigma 2\pi r_i \Delta r_i r_i^2 = 2\pi\sigma \sum_i r_i^3 \Delta r_i$$

$$\sum_i r_i^3 \Delta r_i = \int_0^R r^3 dr = \frac{1}{4}R^4$$

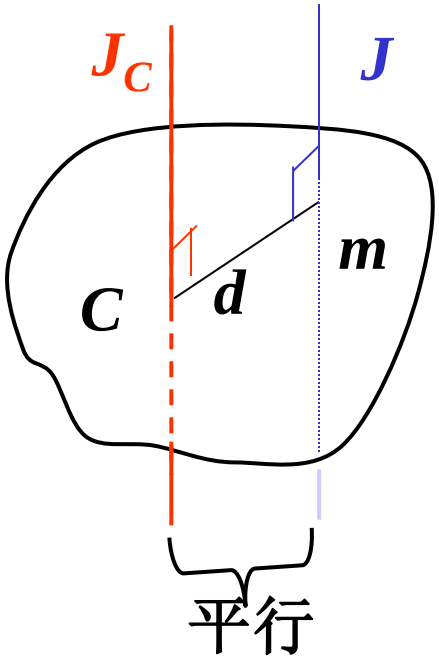
$$J_c = 2\pi\sigma \frac{R^4}{4} = \frac{1}{2}mR^2$$



均匀杆：

$$J_c = \frac{1}{12}ml^2, \quad J_A = \frac{1}{3}ml^2$$

二. 计算 J 的几条规律

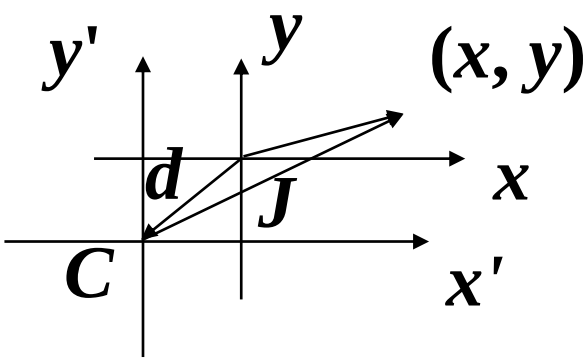


1. 对同一轴 J 具有可叠加性

$$\boxed{J=\sum J_i} \qquad J_z=\sum_i \Delta m_i r_{i\perp}^2$$

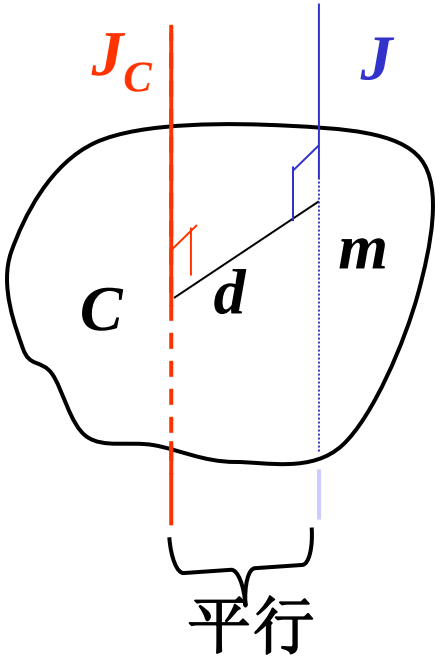
2. 平行轴定理

$$J_z=\sum_i \Delta m_i r_{i\perp}^2=\sum_i \Delta m_i (x_i^2+y_i^2)$$



$$\sum_i \Delta m_i x_i^2=\sum_i \Delta m_i (x_i'+x_c)^2$$

$$=\sum_i \Delta m_i x_i'^2+2x_c\underbrace{\sum_i \Delta m_i x_i'}_{=0}+x_c^2\underbrace{\sum_i \Delta m_i}_{=m}$$

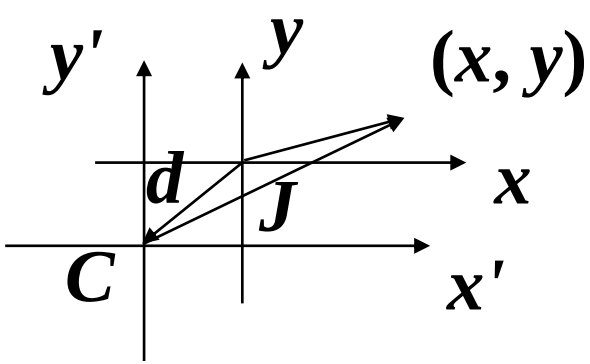


$$J_z = \sum_i \Delta m_i x_i'^2 + 2x_c \sum_i \Delta m_i x_i' + x_c^2 \sum_i \Delta m_i$$

$$+ \sum_i \Delta m_i y_i'^2 + 2x_c \sum_i \Delta m_i y_i' + y_c^2 \sum_i \Delta m_i$$

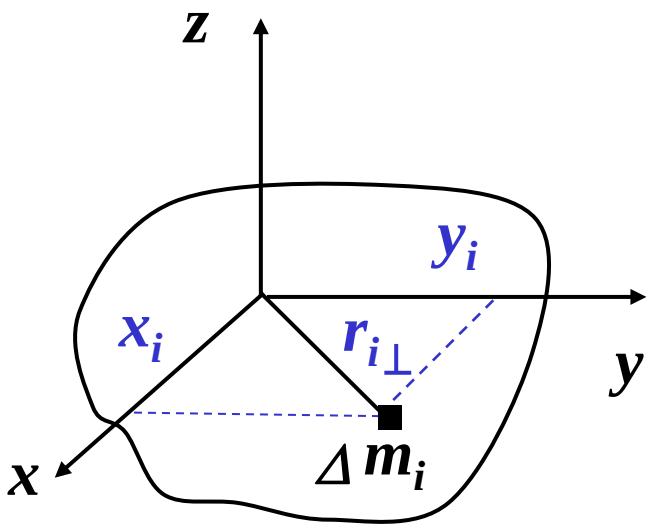
$$x_c^2 + y_c^2 = d^2$$

$$\sum_i \Delta m_i (x_i'^2 + y_i'^2) = J_c$$



$$J = J_c + md^2$$

$$\therefore J_c = J_{\min}$$



3. 对薄平板刚体的正交轴定理

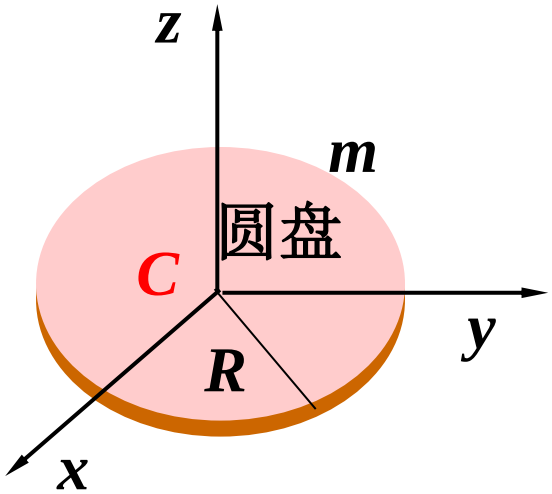
$$J_z = \sum \Delta m_i r_{i\perp}^2$$

$$= \sum \Delta m_i x_i^2 + \sum \Delta m_i y_i^2$$

即 $\boxed{J = J_x + J_y}$

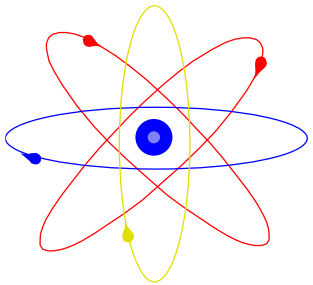
例： 已知圆盘 $J_z = \frac{1}{2} mR^2$

求:对圆盘的一条直径的 J_x（或 J_y）

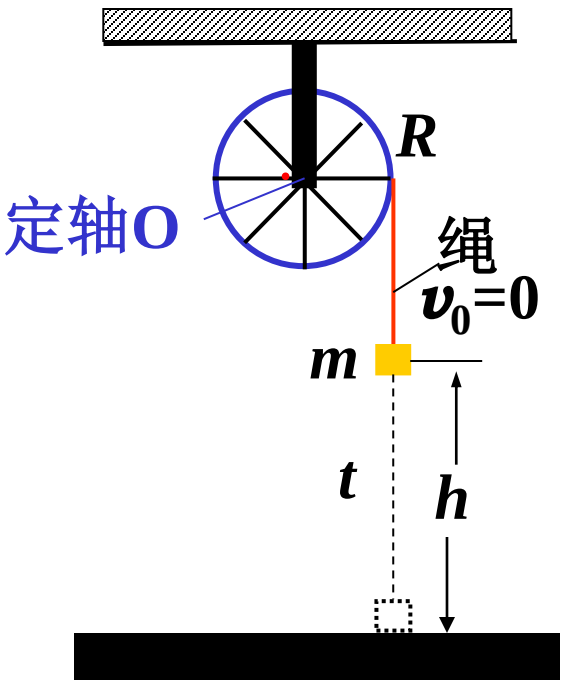


由 $\begin{cases} J_z = J_y + J_x \\ J_x = J_y \end{cases}$

$$\therefore \quad J_x = J_y = \frac{1}{4} mR^2$$



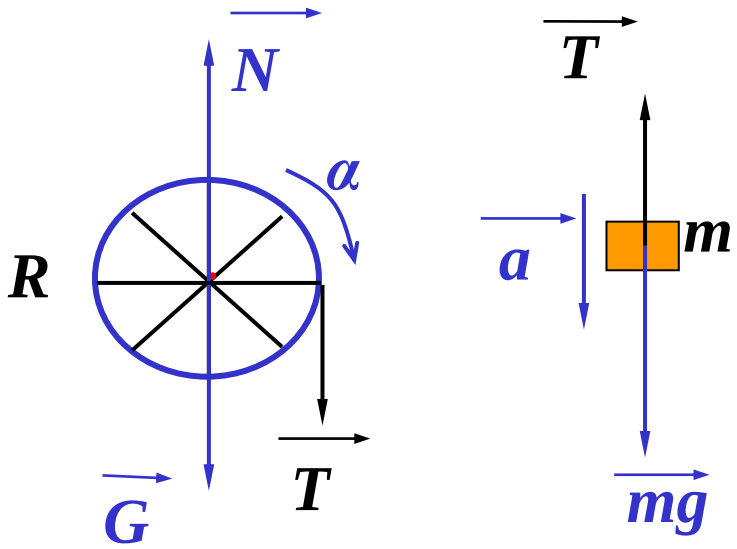
§ 5.4 刚体定轴转动定律的应用



已知： R =0.20m， m =1.0kg , v_0= 0，
h =1.5 m，绳轮无相对滑动，绳不可伸长，下落时间
t =3.0 s

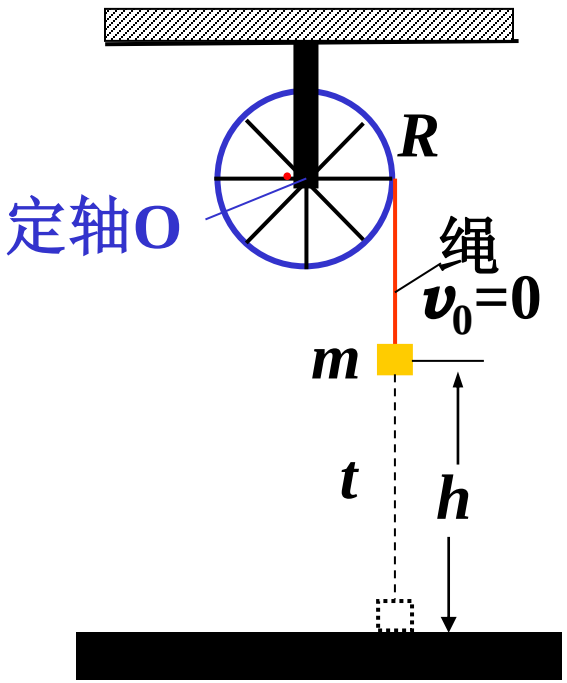
求：轮对 O 轴 J=？

解：动力学关系：



对轮： $TR = J\alpha$ (1),

对 m： $mg - T = ma$ (2)



$$mg - J\frac{\alpha}{R} = ma$$

运动学关系： $\alpha = \dfrac{a}{R}$ (3)

$$a = \frac{g}{1+\dfrac{J}{mR^2}}$$

$$h = \frac{1}{2}at^2$$

联立解得： $J = (\dfrac{gt^2}{2h} - 1)mR^2$

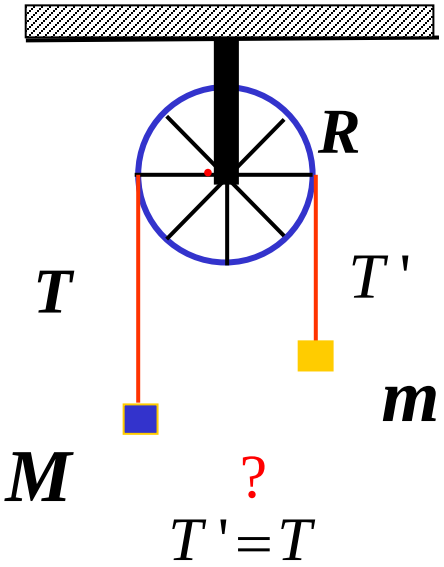
$$= (\frac{9.8\times 3.0^2}{2\times 1.5} - 1)\times 1.0\times 0.20^2 = 1.1\,\text{kg m}^2$$

注意有效数字

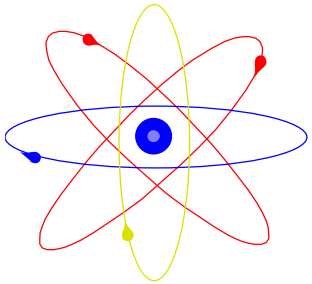
分析： 1. 单位对；

2. h、m一定， $J \rightarrow t$ ，合理；

3. 若 $J = 0$，得 $h = \frac{1}{2}gt^2$，正确。

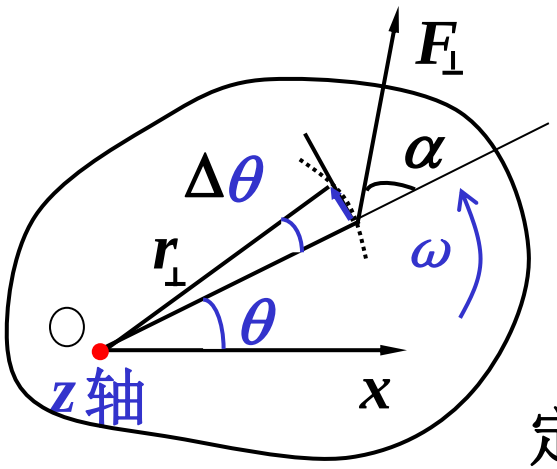


运动方程?



§ 5.5 定轴转动中的功和能

一. 力矩的功



$$\Delta W = F_\perp \sin\alpha\,(r_\perp \Delta\theta)$$
$$= F_\perp r_\perp \sin\alpha\,\Delta\theta$$
$$= M\Delta\theta$$

$$\boxed{\Delta W = M\Delta\theta}$$

定轴转动元功 = 沿z轴力矩×绕z轴小转角

$$\Delta W = F\Delta s$$

$W=\sum_i M_i\Delta\theta_i$ —力矩的空间角度积累效应

二. 定轴转动动能定理

$$M_{外z} = \frac{dL_z}{dt} = J_z \frac{d\omega}{dt}$$

类比质点一维情形： $F = m\frac{dv}{dt}$

$$W = \frac{1}{2}J\omega_2^2 - \frac{1}{2}J\omega_1^2$$

J --> m $\omega = \frac{d\theta}{dt}$ --> $v = \frac{ds}{dt}$

$$W = E_{k2} - E_{k1}$$

$$\therefore \boxed{E_k = \frac{1}{2}J\omega^2}$$ —— 转动动能

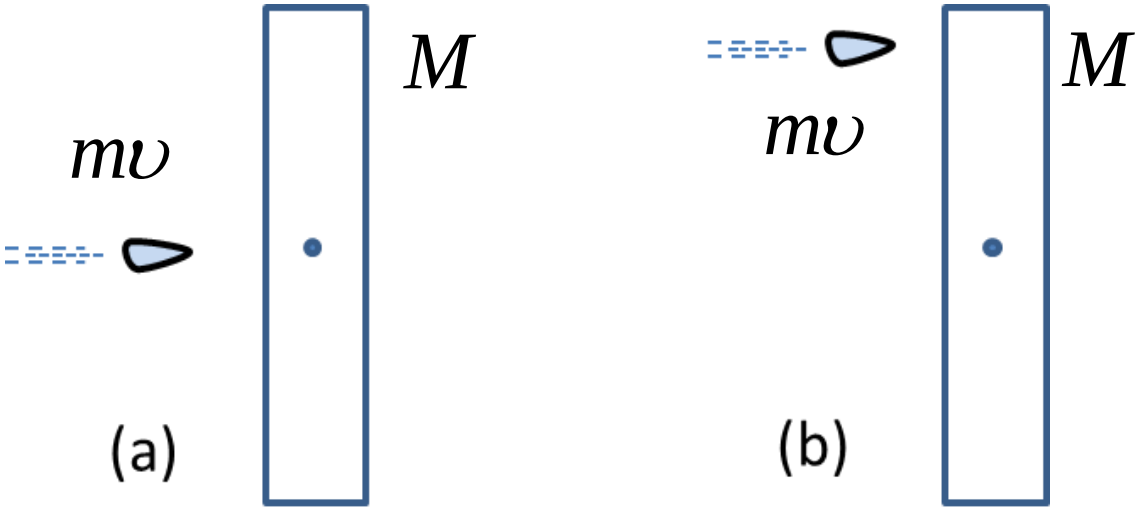
$$\frac{1}{2}J\omega^2 = \frac{1}{2}\sum_i \Delta m_i r_{\perp i}^2 \omega^2 = \frac{1}{2}\sum_i \Delta m_i \upsilon_i^2$$

与质点系比较-----刚体质点间内力不做功

单选题 1分

设置

有个均匀木棒平放在光滑的水平面上，一子弹水平射入其中，如图所示的（a）和（b）两种情况，哪次木棒获得的动能更大？



A (a)大

B (b)大

C 一样大

提交

三. 定轴转动的功能原理

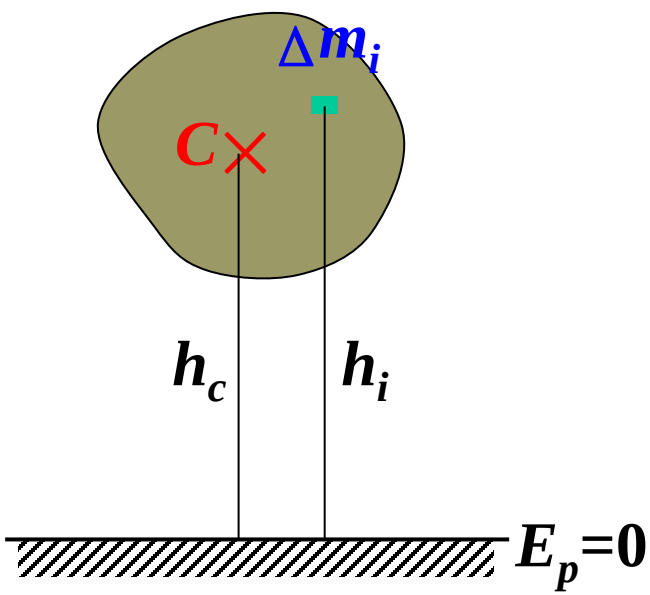
质点系功能原理对刚体仍成立：

$$W_{外} + \underbrace{W_{内非}}_{\to 0} = (E_{k2} + E_{p2}) - (E_{k1} + E_{p1})$$

若 $W_{外} = 0$，

则 $E_k + E_p =$ 常量

刚体重力势能：



$$E_p = \sum \Delta m_i g h_i$$

$$= mg \frac{\sum \Delta m_i h_i}{m} = mgh_c$$

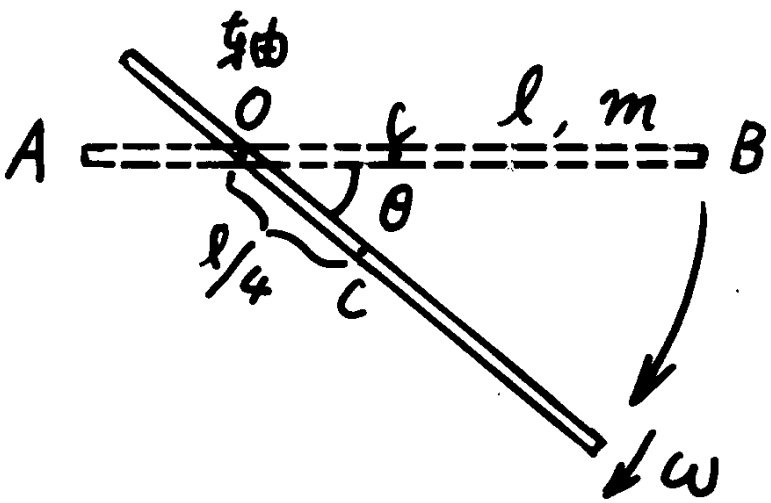
所有质量集中于质心时的势能

[例]已知：均匀直杆 m，长为 l，初始水平静止，轴光滑，

$\overline{AO}=\dfrac{l}{4}$ 。 求 ：杆下摆 θ 角后，角速度 ω =?

轴对杆作用力 $\vec{N}$ =?

解：杆+地球系统， ∵只有重力作功，∴ E 守恒



初始： $E_{k1}=0$，

令 $E_{P1}=0$

末态： $E_{k2}=\dfrac{1}{2}J_o\omega^2$，

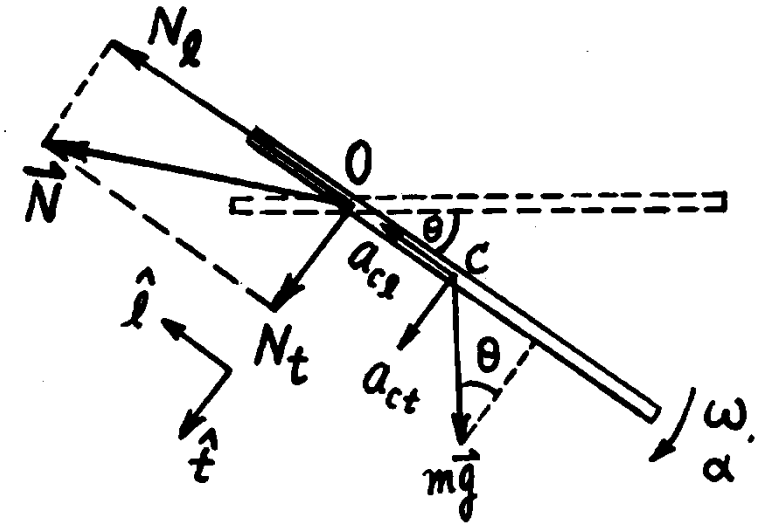
$$E_{P2}=-mg\frac{l}{4}\sin\theta$$

则： $\dfrac{1}{2}J_o\omega^2-mg\dfrac{l}{4}\sin\theta=0$ (1)

由平行轴定理

$$J_o = J_c + md^2 = \frac{1}{12}ml^2 + m(\frac{l}{4})^2$$

$$= \frac{7}{48}ml^2 \qquad (2)$$



由(1)、(2)得： $\omega = 2\sqrt{\dfrac{6g\sin\theta}{7l}}$

定轴转动定律

$$mg\frac{l}{4}\cos\theta = J_o\alpha$$

质心运动定理： $\vec{N} + m\vec{g} = m\vec{a}_c$

$\hat{l}$方向： $-mg\sin\theta + N_l = ma_{cl}$ (3)

$\hat{t}$方向： $mg\cos\theta + N_t = ma_{ct}$ (4)

$$a_{cl} = \frac{l}{4}\omega^2 = \frac{6}{7} g \sin\theta \qquad (5)$$

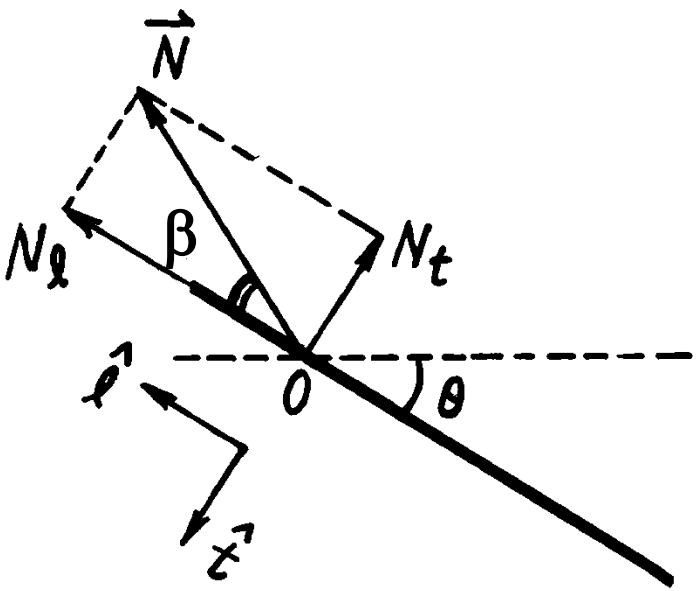
$$a_{ct} = \frac{l}{4}\alpha = \frac{l}{4}\,\frac{\frac{l}{4} mg \cos\theta}{J_o} = \frac{3g\cos\theta}{7} \qquad (6)$$

由(3)(4)(5)(6) 可解得：

$$N_l = \frac{13}{7} mg \sin\theta,$$

$$N_t = -\frac{4}{7} mg \cos\theta$$

$$\vec{N} = \frac{13}{7} mg\sin\theta\ \hat{l} - \frac{4}{7} mg\cos\theta\ \hat{t}$$



§ 5.6 刚体定轴转动的角动量定理和角动量守恒定律

现在讨论力矩对时间的积累效应：冲量矩

质点系：

对点：$\boxed{\vec{M}_{外} = \dfrac{d\vec{L}}{dt}}$，

$$\boxed{\int_{t_1}^{t_2} \vec{M}_{外} dt = \vec{L}_2 - \vec{L}_1}$$

对轴：$\boxed{M_{外z} = \dfrac{dL_z}{dt}}$，

$$\boxed{\int_{t_1}^{t_2} M_{外z} dt = L_{z2} - L_{z1}}$$

定轴转动刚体：$L_Z = J_Z \omega$

$$\int_{t_1}^{t_2} M_{外z} dt = J_z\omega_2 - J_z\omega_1$$ —— 刚体定轴转动的角动量定理

当 $M_{外z} = 0$ 时，$J_z \omega = \text{const}$.

角动量守恒

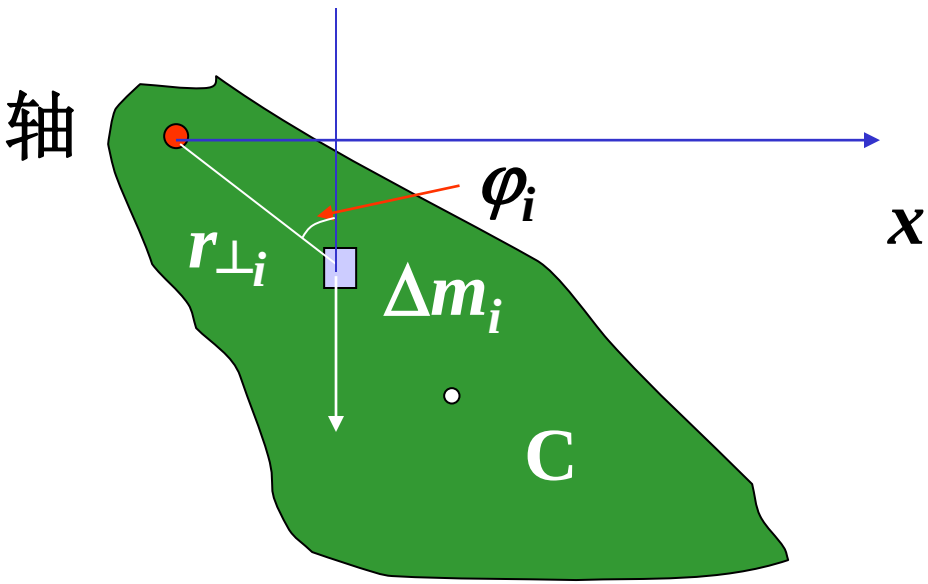
$\begin{cases} \text{大小不变} \\ \text{正、负不变} \end{cases}$

若刚体由几部分组成，且都绕同一轴转动

当 $M_{外z}=0$ 时， $\sum J_{iz}\omega_i=\text{const.}$ ，这时角动量可在内部传递

重力力矩

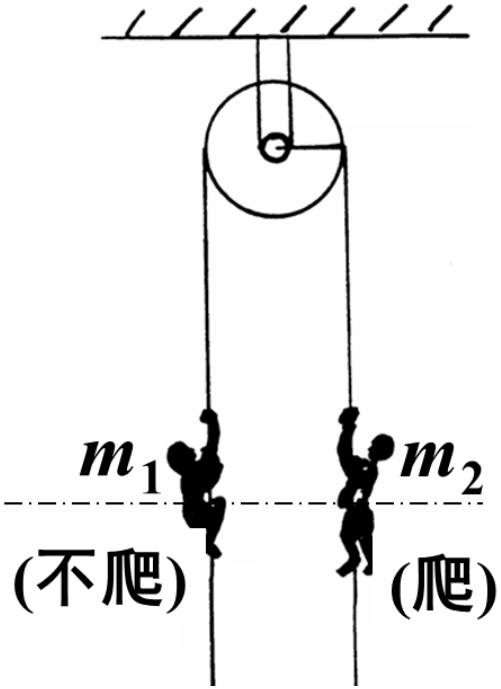
$$M=\sum M_i=\sum r_{\perp i}\Delta m_i g\sin\varphi_i$$



$$M=\sum x_i\Delta m_i g=x_c mg$$

好像所有物质集中于质心

例 如图示，两个同样重的小孩，各抓着跨过滑轮的一端。起初都不动，然后右边的小孩用力向上爬绳，左边的小孩只抓住绳但不爬动。忽略绳的质量和轴的摩擦，但滑轮的质量不能忽略。设他们的起始高度相同，问哪个小孩先到达滑轮？



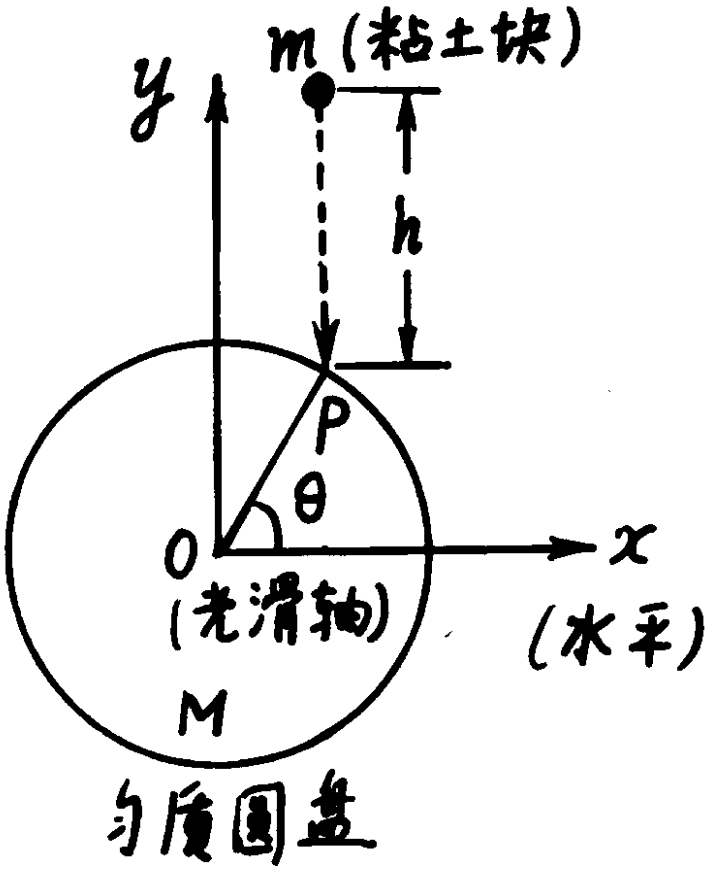
解答：小孩重力相等，相对轴心力矩大小相等方向相反，那么整个系统合外力矩为零，**角动量守恒**。

相对于轴心，

$$mv_1r + J\omega = mv_2r$$

$$v_1 < v_2$$

[例]如图示已知： $M = 2m$，h，$\theta = 60^\circ$



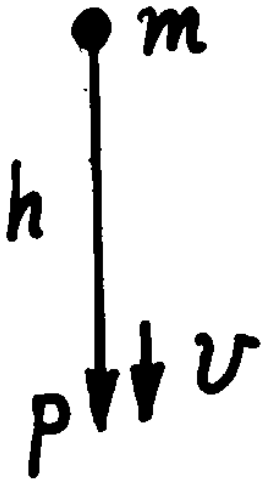
求：碰撞后瞬间盘的 $\omega_0 =$ ？

P 转到 x 轴时盘的 $\omega =$? $\alpha =$?

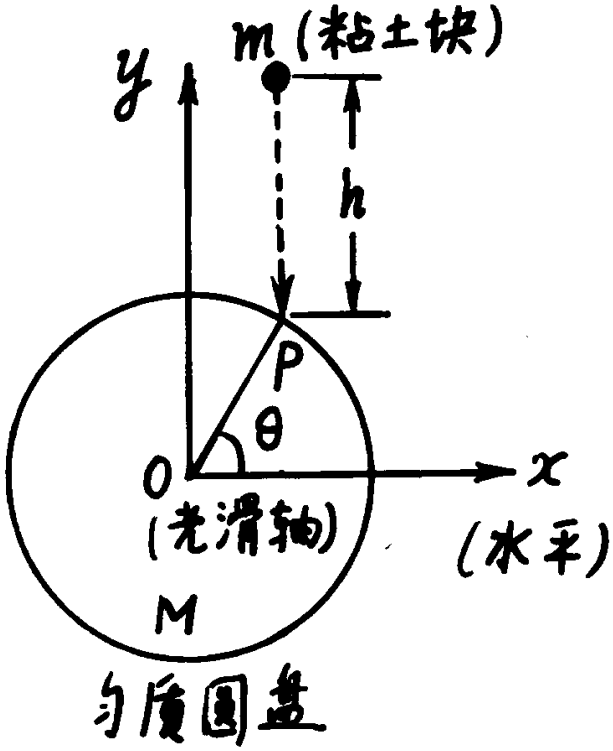
解： m下落：

$$mgh = \frac{1}{2}mv^2$$

$$\Rightarrow v = \sqrt{2gh} \quad (1)$$



碰撞 Δt 极小，对 m +盘系统，冲力远大于重力，故重力对O力矩可忽略，角动量守恒：



$$mvR\cos\theta = J\omega_o \qquad (2)$$

$$J = \frac{1}{2}MR^2 + mR^2 = 2mR^2 \qquad (3)$$

由 (1)(2)(3)得：

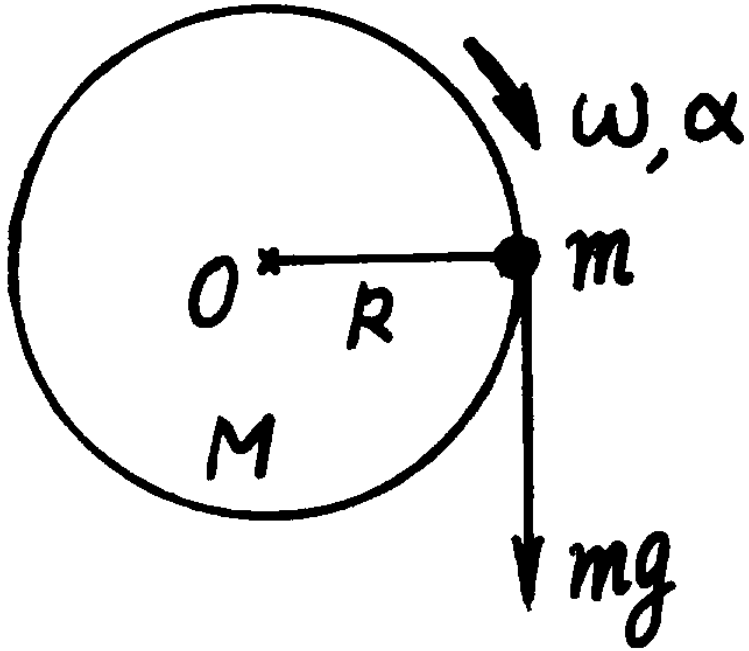
$$\omega_o = \frac{\sqrt{2gh}}{2R}\cos\theta \qquad (4)$$

对 m+M+地球系统，E 守恒，

令 P 与x轴 重合时 E_P=0。

则：

$$mgR\sin\theta + \frac{1}{2}J\omega_o^2 = \frac{1}{2}J\omega^2 \qquad (5)$$



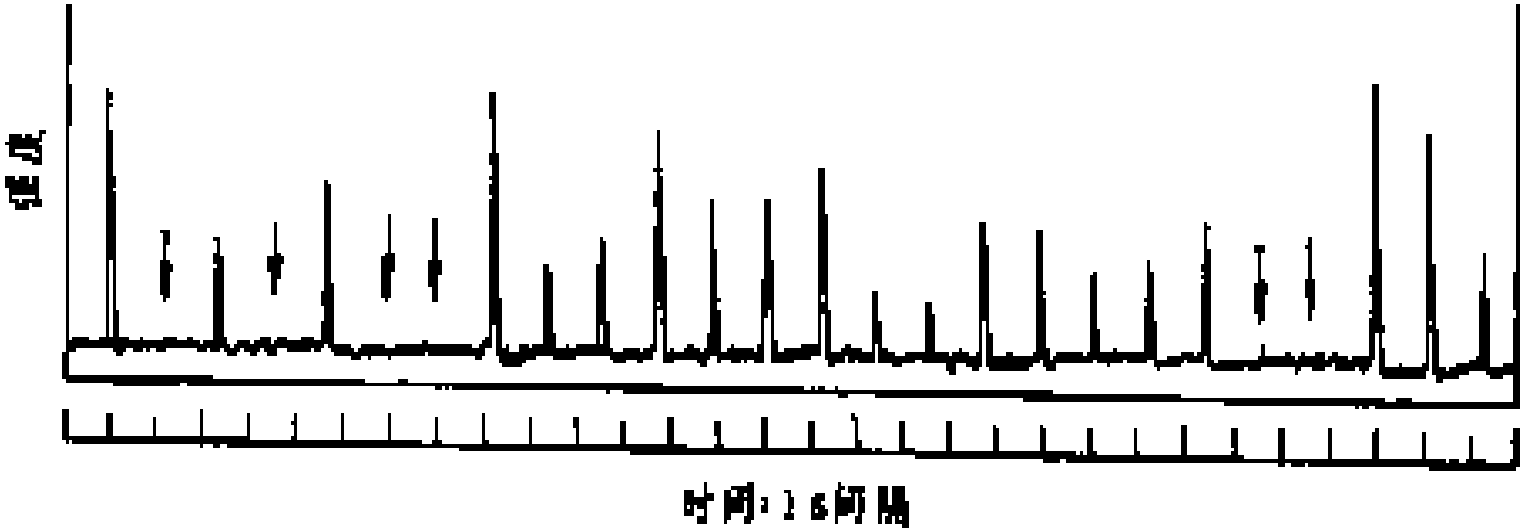
由 (3)(4)(5)　得：

$$\omega = \sqrt{\frac{gh}{2R^2}\cos^2\theta + \frac{g}{R}\sin\theta}$$

$$= \frac{1}{2R}\cdot\sqrt{\frac{g}{2}(h + 4\sqrt{3}R)}$$

$(\theta = 60°)$

$$\alpha = \frac{M}{J} = \frac{mgR}{2mR^2} = \frac{g}{2R}$$



脉冲星的精确周期性信号

脉冲星自转周期几乎不变，
转速太快，应为中子星（密度太小则被离心力撕裂）

球形星体自转角动量 $= \frac{2}{5} MR^2 \omega$

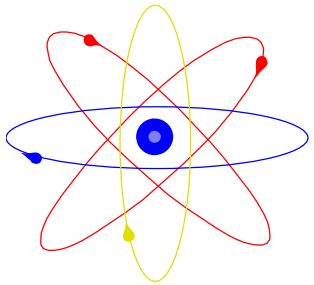
星体演化过程不受外力矩，角动量守恒

*上图中的脉冲星自转周期只有约 **1.19** 秒，要使星体不被惯性离心力甩散，必须满足条件：

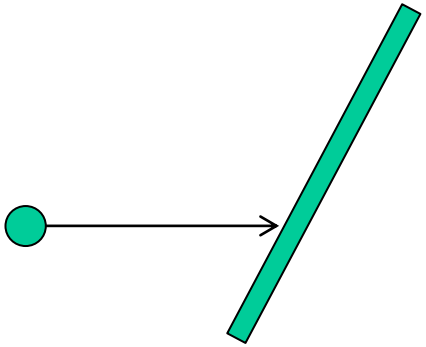
$$\frac{GM}{R^2} > R\omega^2 \text{，} \left(M = \frac{4\pi}{3}R^3\rho\right)$$

即星体的密度需满足条件：$\rho > \dfrac{3\omega^2}{4\pi G}$

按上条件计算，脉冲星密度超过了白矮星密度。经多方认证，脉冲星是高速旋转的中子星。通常中子星自转周期是毫秒量级。



§ 5.7 *刚体的平面运动

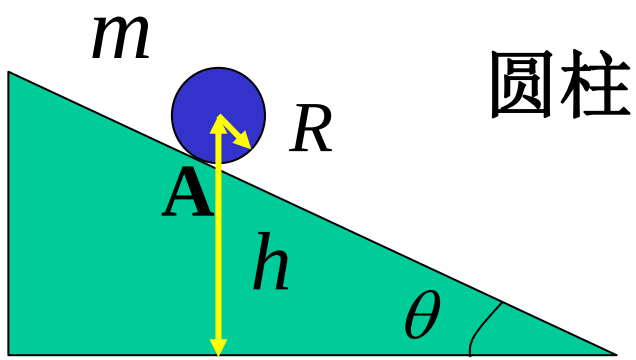


“转轴”不固定, 平移

考虑通过质心的转轴（垂直于平面）

质心平动	$\vec{F} = \dfrac{d\vec{P}_c}{dt} = m\dot{\vec{\upsilon}}_c$
绕质心转动	$\vec{M}' = \dot{\vec{L}}'$ $\quad M' = \dot{L}'$

例



圆柱

摩擦力大, 纯滚动

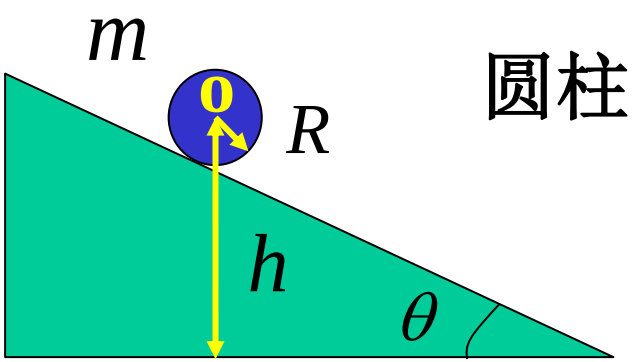
求: 速度和加速度

解: 1. 质心运动

$$mg\sin\theta - f_r = ma_c$$

绕质心转动 $\quad f_r R = J_o \alpha \qquad a_c = R\alpha$

$$a_c = \frac{g\sin\theta}{1+\dfrac{J_o}{mR^2}} = \frac{2}{3}g\sin\theta$$



圆柱

2.机械能守恒

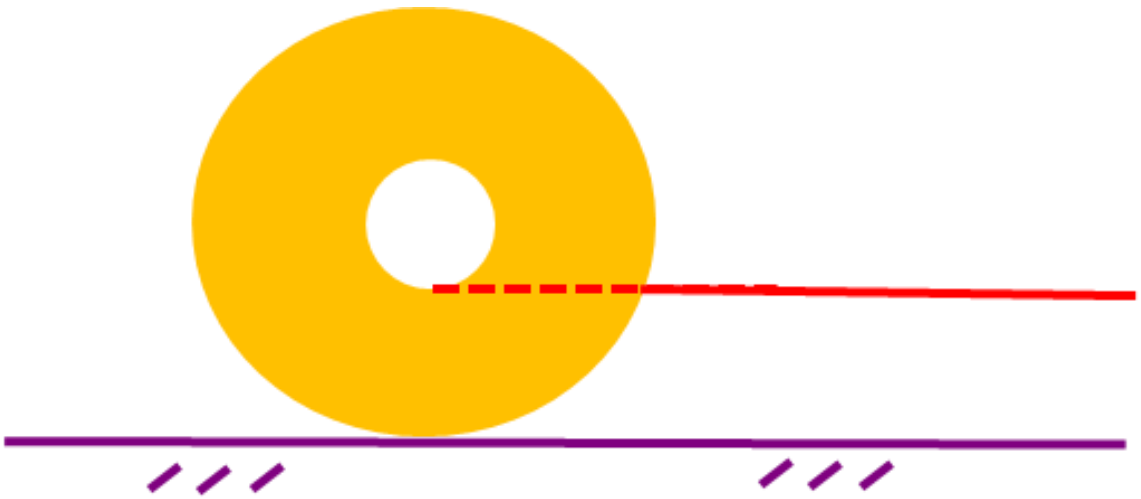
$$mgh = \frac{1}{2}mv_c^2 + \frac{1}{2}J_o\omega^2 \qquad v_c = R\omega$$

$$v_c = 2\sqrt{\frac{gh}{3}} \qquad a_c = \dot{v}_c = \frac{2}{3}g\sin\theta$$

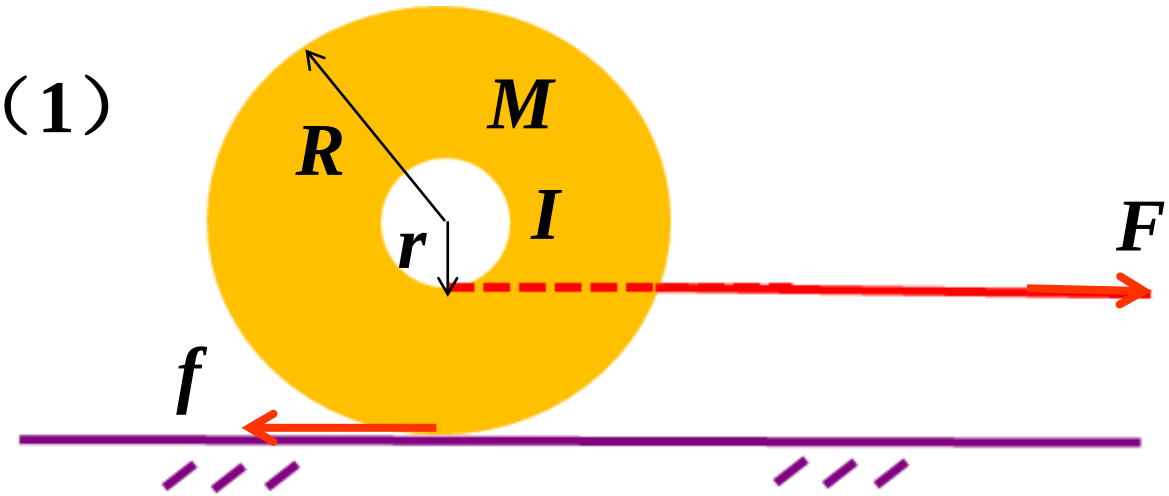
例 在粗糙的水平面上放置一轮子，绳子绕在中轴（与轮子一体）上，如图所示（对着纸面看，绳子是逆时针方向绕在中轴上）。由于绳子绕在轮子的中轴上，拉绳子就会带动轮子滚动。问：

（1）小心地向前慢拉绳子，使轮子不打滑，轮子是向前还是向后滚动？

（2）向前用力快拉绳子，轮子是向前还是向后移动？



解 （1）



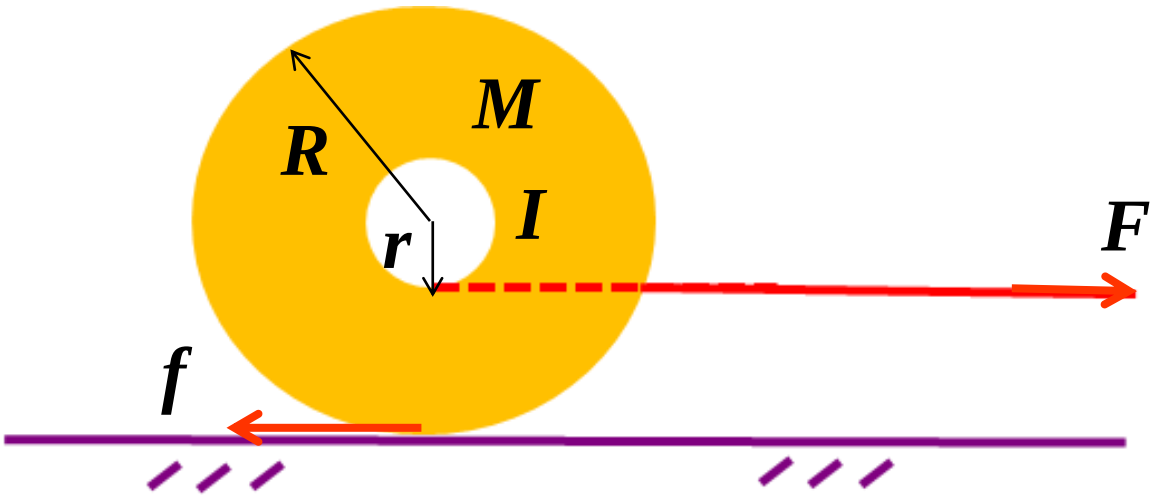
质心运动方程 $F - f = Ma$

绕质心转动定律 $-Fr + fR = I\alpha$

无滑动 $a = R\alpha$

解得 $\alpha = \dfrac{F(R-r)}{MR^2 + I}$ 轮子向前滚动

(2)



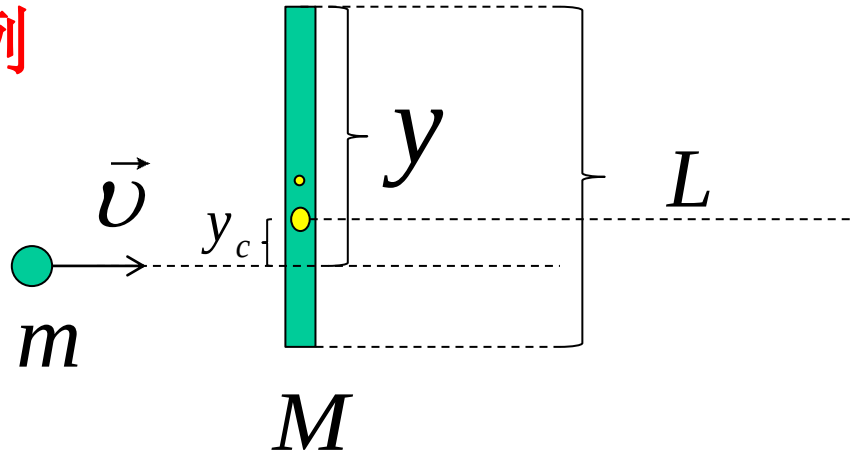
质心运动方程 $F - f = Ma$

$a > 0$ 轮子向前

绕质心转动定律 $-Fr + fR = I\alpha$

$\alpha < 0$ 轮子向后转动，打滑

例



平面上
完全非弹性碰撞, 求:
碰后细杆儿的角速度

解:

碰撞点为原点,
碰撞瞬间系统质心

$$x_c = 0$$

$$y_c = \frac{M}{m+M}(y - \frac{L}{2})$$

质心速度 $$V_c = \frac{m\upsilon}{m+M}$$

质心系

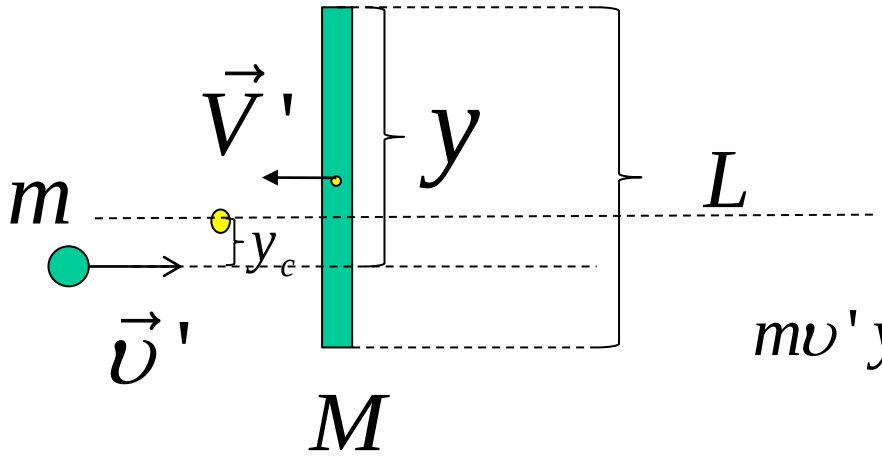
碰撞前

质点速度

$$\upsilon' = \frac{M\upsilon}{m+M}$$

杆中心速度

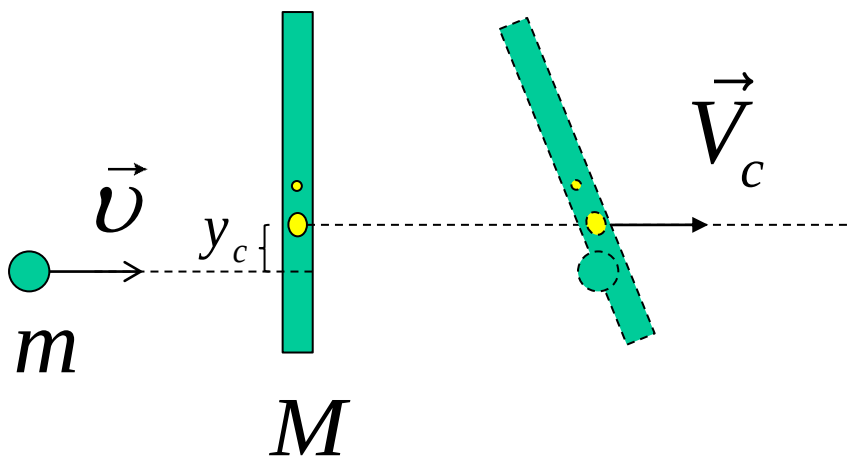
$$V' = -\frac{m\upsilon}{m+M}$$



在质心系角动量守恒

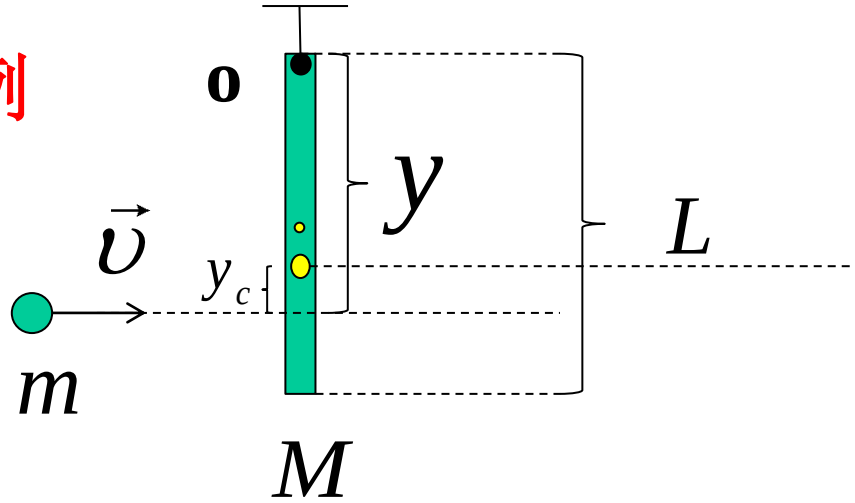
$$m\upsilon' y_c + M(-V')(y - \frac{L}{2} - y_c) = J_c\omega$$

$$J_c = my_c^2 + M(y - \frac{L}{2} - y_c)^2 + \frac{1}{12}ML^2 = \frac{mM}{m+M}(y - \frac{L}{2})^2 + \frac{1}{12}ML^2$$



$$\omega = \frac{mM}{m+M}\frac{\upsilon(y - \frac{L}{2})}{J_c}$$

例



悬挂杆
完全非弹性碰撞,
碰撞时间 Δt
求: 1) 碰撞瞬间转轴受平均力
2) 杆摆最大角

解:

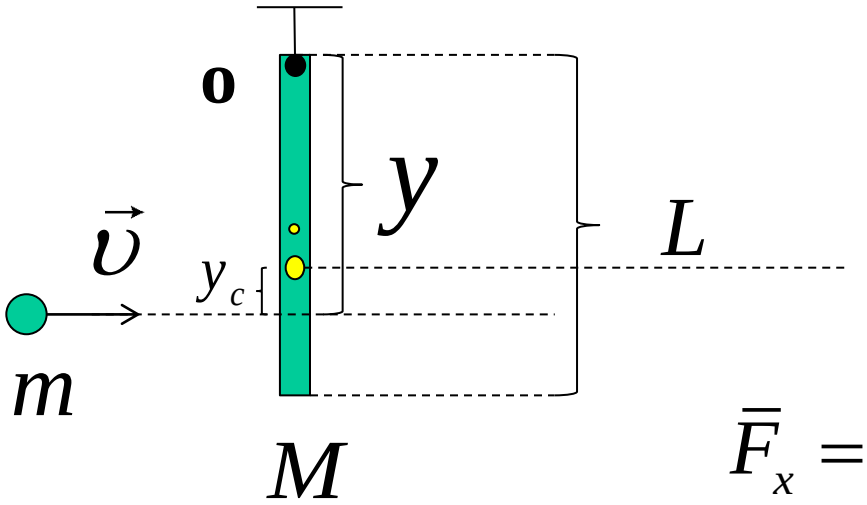
1）在质心系 $\vec{M}' = \dot{\vec{L}}'$ $\vec{M}'\Delta t = \Delta\vec{L}'$

$$\bar{F}_x(y - y_c)\Delta t = -J_c\omega - (-my_c\upsilon)$$

$$y_c = \frac{M}{m+M}(y - \frac{L}{2})$$

绕固定点 o 角动量守恒

$$m\upsilon y = J_o\omega$$



$$\bar{F}_x = \frac{m\upsilon(y_c - \frac{J_c}{J_o}y)}{(y - y_c)\Delta t}$$

$$\bar{F}_x = -\frac{m\upsilon}{2\Delta t}\frac{ML}{my^2 + \frac{1}{3}ML^2}(y - \frac{2}{3}L)$$

$$F_y = (M\frac{L}{2} + my)\omega^2 + (m+M)g \qquad \bar{F}_y = ?$$

$$Mg \rightarrow (M\frac{L}{2} + my)\omega^2 + (m+M)g$$

2) 杆摆最大角

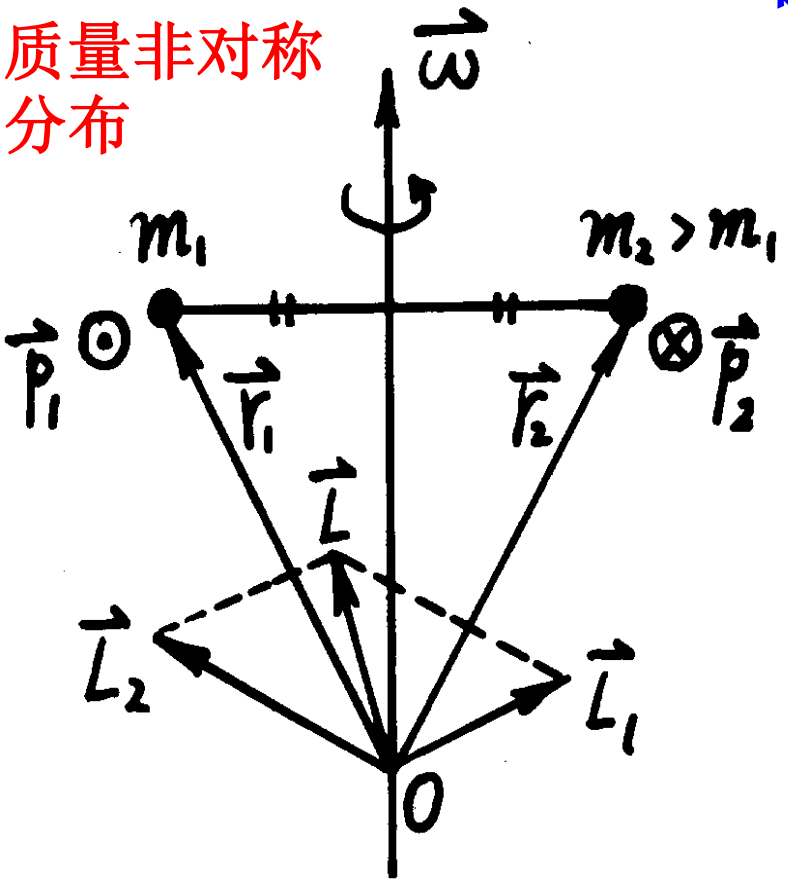
$$\frac{1}{2}J_o\omega^2 = (M+m)g(y - y_c)(1 - \cos\theta)$$

§ 5.8 *进动（Precession）(定点转动的特例）

定轴转动 $L_z\hat{z} \,//\, \vec{\omega}$（对轴）

但对定点转动 $\vec{L}$ 还平行于 $\vec{\omega}$ 吗？

质量非对称分布



质量分布对称、且刚体绕对称轴转动时，对轴上任一点有：

$$\vec{L} \,//\, \vec{\omega} \,//\, 轴$$

$$\vec{L}(对点) = L_z\hat{z}(对轴) = J_z\vec{\omega}$$

显然若 $m_1 \neq m_2$ (不对称)，

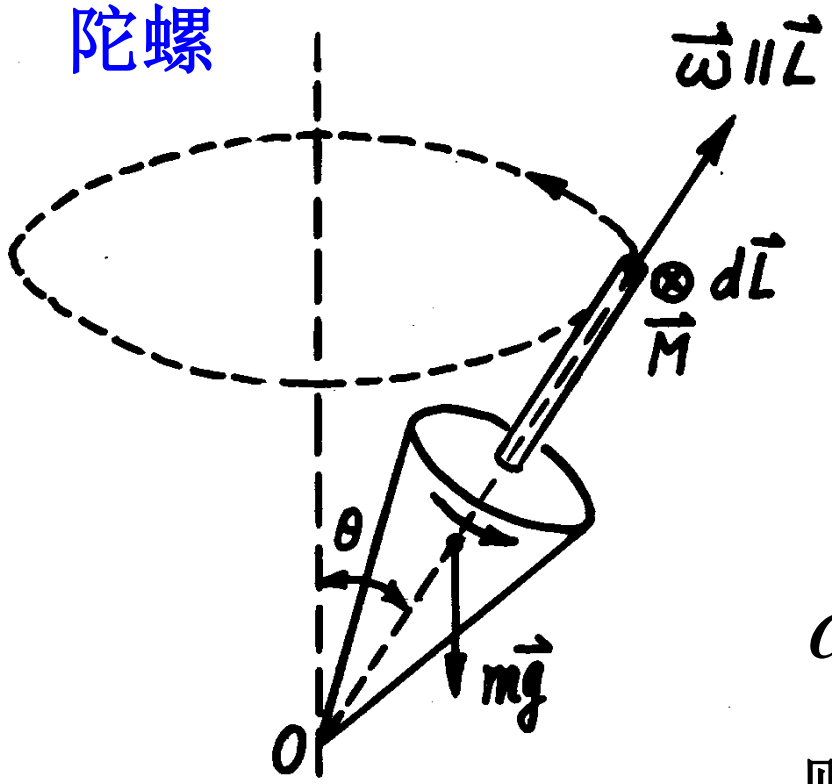
则 $\vec{L} \not\parallel \vec{\omega}$ （对 0 点），

角动量 $\vec{L}$ 做进动 轴受力矩

旋进：高速旋转的物体，其自转轴绕另一个轴转动的现象。

讨论具有对称轴的刚体受力矩进动问题

陀螺



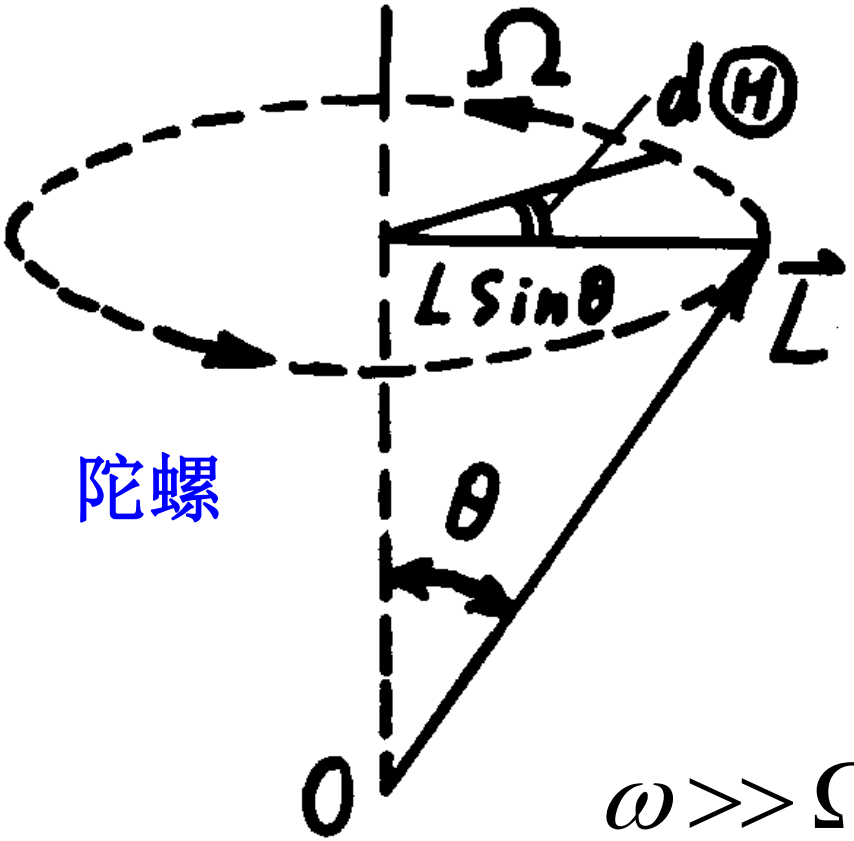
$$\vec{M} = \frac{d\vec{L}}{dt}$$

$$d\vec{L} = \vec{M}dt \,//\, \vec{M}$$

当 $\vec{M} \perp \vec{L}$ 时， $d\vec{L} \perp \vec{L}$

$$\omega >> \Omega$$

则 $\vec{L}$ 只改变方向 不改变大小。



陀螺

$$\omega >> \Omega$$

$$\dot{\vec{L}} \approx \vec{\Omega} \times \vec{L}$$

令 $$\Omega = \frac{d\Theta}{dt}$$

$$\left|d\vec{L}\right| = L\sin\theta \; d\Theta$$

则 $$M = \frac{\left|d\vec{L}\right|}{dt} = \frac{L\sin\theta\, d\Theta}{dt}$$

$$= L\sin\theta \; \Omega$$

$$\therefore \quad \Omega = \frac{M}{L\sin\theta} = \frac{M}{J\omega\sin\theta} = \frac{mgl_c}{J\omega}$$

以上只是近似讨论，因为当旋进发生后：

$$\vec{\omega}_{总} = \vec{\omega} + \vec{\Omega}$$

只有高速自转 $\omega >> \Omega$ 时，才有 $\vec{\omega}_{总} \doteq \vec{\omega}$

这时才有 $\vec{L} \doteq J\vec{\omega}$

当考虑到 $\vec{\Omega}$ 对 $\vec{\omega}_{总}$ 的贡献时，

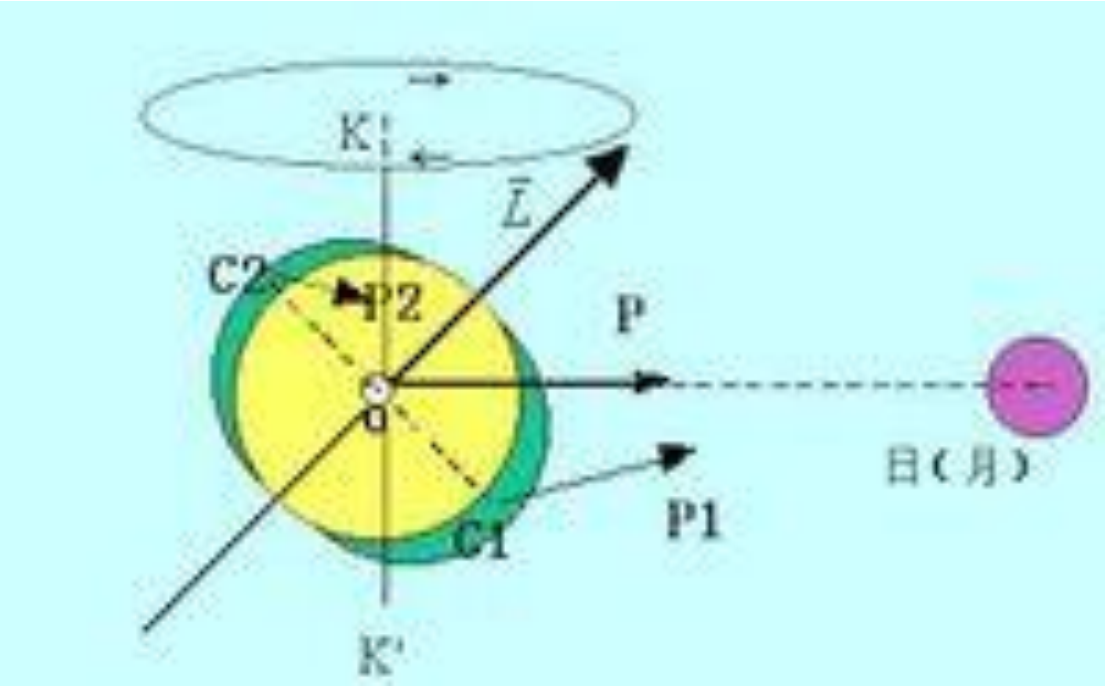
自转轴在旋进时, 有时还会出现

微小的上下的周期摆动， 这种运动叫章动 (**nutation**)。

▲地球转轴的进动：非球效应，进动周期　25800 年

3000 年前，周，北极是 β，现 α，

12000 年后，织女星。



第五章完

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